Homework #8

This is the homework for week 10. The due date is 11:59 p.m., November 19. Late submission will automatically result in "0" grade. Please include codes for the program part in your submission.

1 Computation Questions

(**Question1**): (40 points) Let $X_1, ..., X_n \overset{iid}{\sim} Unif(0, \theta)$ and let $Y = \max\{X_1, ..., X_n\}$. We want to test $H_0 : \theta = 1/2$ v.s. $H_1 : \theta > 1/2$.

Suppose we decide to test this hypothesis by rejecting H_0 when $Y > c$.

(a) Find the power function.

(b) What choice of c will make the size (i.e., Type-I error) of the test 0.05?

(c) In a sample of size $n = 20$ with $Y = 0.48$ what is the p-value? What conclusion about H_0 would you make?

(d) In a sample of size $n = 20$ with $Y = 0.52$ what is the p-value? What conclusion about H_0 would you make?

(**Question2**): (20 points) There is a theory that people can postpone their death until after an important event. To test the theory, Phillips and King (1988) collected data on deaths around the Jewish holiday Passover. Of 1919 deaths, 922 died the week before the holiday and 997 died the week after. Think of this as a binomial and test the null hypothesis that $\theta = 1/2$.

(a) Report and interpret the p-value.

(b) Construct a confidence interval for θ.

Phillips, D. and King, E. (1988). Death takes a holiday: Mortality surrounding major social occasions. *Lancet* 2 728–732.

(**Question3**): (20 points) Let $X \sim Binomial(n, p)$.

(a) Construct the likelihood ratio test for

$$H_0 : p = p_0 \text{ v.s. } H_1 : p \neq p_0.$$

(b) Compare to the Wald test (based on Normal approximation).

2 Programming Questions

(20 points)

Implement the following experiment using Jupyter or R:

Let $Y_1, Y_2, ..., Y_n$ be n independent random variables with normal distributions $N(\beta x_i, \sigma^2)$, where $x_1, x_2, ..., x_n$ are known and β and σ^2 are unknown parameters.

Generate synthetic data using random seed 1, $\beta = 0$, $\sigma = 2$ ($\sigma^2 = 4$) and the following x_i's

3.4, 3.6, 3.8, 3.3, 3.4, 3.5, 3.7, 3.6, 3.7.

(1) Find the likelihood ratio test for $H_0 : \beta = 0$ v.s. $H_1 : \beta \neq 0$.

(2) What is your conclusion?